



RUHR-UNIVERSITÄT BOCHUM

Valence Asymmetries – A Perspective from Cognitive Psychology

Hans Alves | Valence Asymmetries Meeting UPF 26

A Personal Chronology of Valence Asymmetry Research

- How it started for me - similarity and processing speed
- Extension to recognition memory
- How general is the similarity asymmetry?
- From Similarity to Diversity
- Explanations
- The Common Good Phenomenon
- Why are most stereotypes negative?
- Intergroup Biases

How it Started for me

Journal of Personality and Social Psychology
2008, Vol. 95, No. 1, 36–49

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0022-3514/08/\$12.00 DOI: 10.1037/0022-3514.95.1.36

Why Positive Information Is Processed Faster: The Density Hypothesis

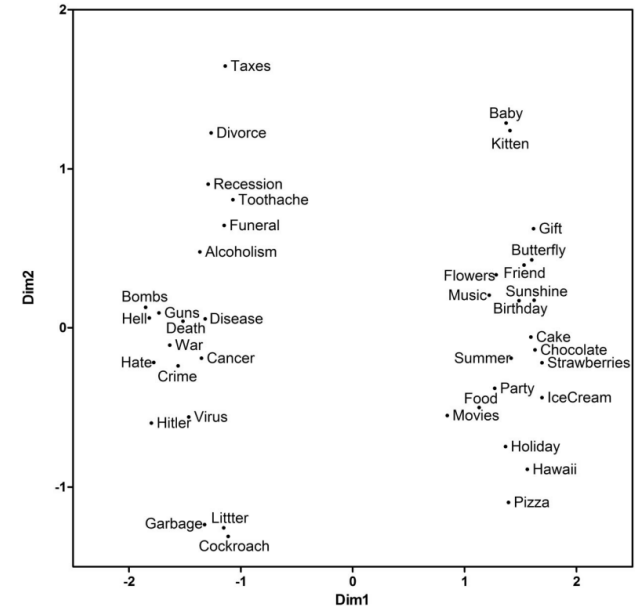
Christian Unkelbach, Klaus Fiedler, Myriam Bayer, Martin Stegmüller, and Daniel Danner
Universität Heidelberg

Processing Speed

Evaluative Priming

Smile -> Cake -> positive (faster)

Divorce -> Cockroach -> negative (slower)



Extension to Recognition Memory

A density explanation of valence asymmetries in recognition memory

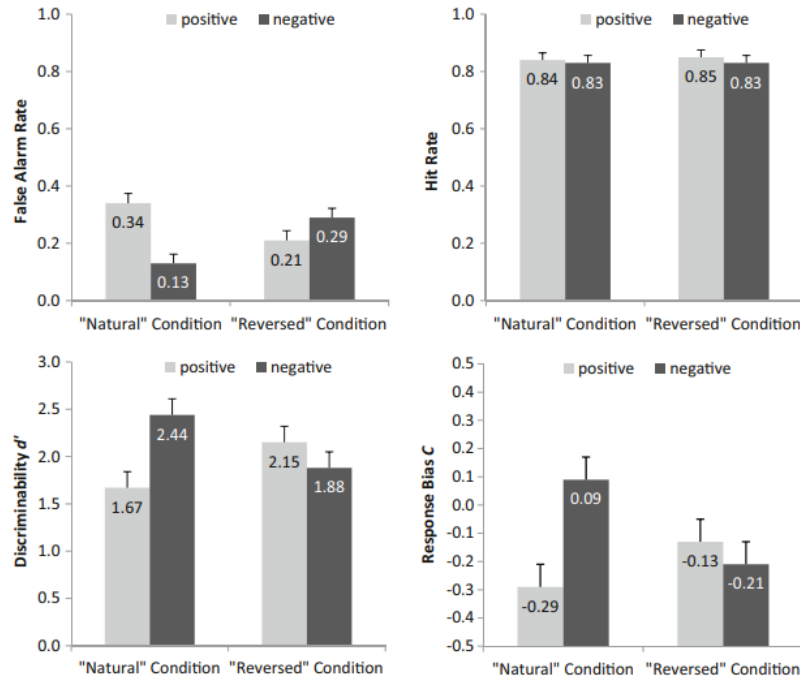
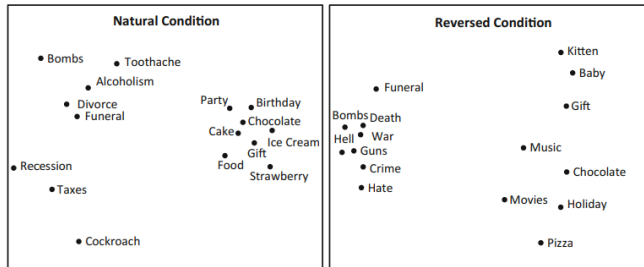
Hans Alves · Christian Unkelbach · Juliane Burghardt · Alex S. Koch · Tobias Krüger · Vaughn D. Becker

Typical findings

Higher **accuracy** for negative stimuli

Higher **response bias** („old“) for positive stimuli

-> Explained by **higher similarity** among positive stimuli



How General is the Similarity Asymmetry?

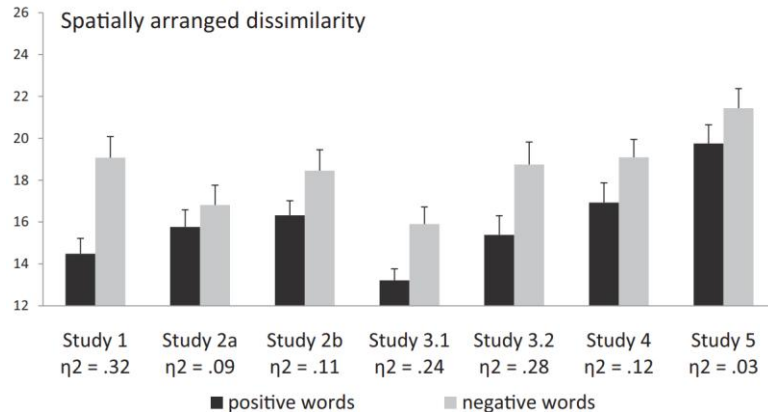
Journal of Experimental Psychology:
Learning, Memory, and Cognition
2006, Vol. 32, No. 8, 1171–1192

© 2006 American Psychological Association
0278-7393/06/\$12.00 http://dx.doi.org/10.1037/0278-7393.32.8.1171

Problem of pairwise similarity ratings

50 stimuli result in 1225 pairwise comparisons

New Method: SPAM – Spatial Arrangement Method



Study 1: words of extreme valence
(see Unkelbach et al., 2008)

Study 2a: self-selected words

Study 2b: words selected by
another participant

Study 3.1: self-selected words
of consensual valence

Study 3.2: self-selected words
of idiosyncratic valence

Study 4: words selected by many
others; no mention of valence

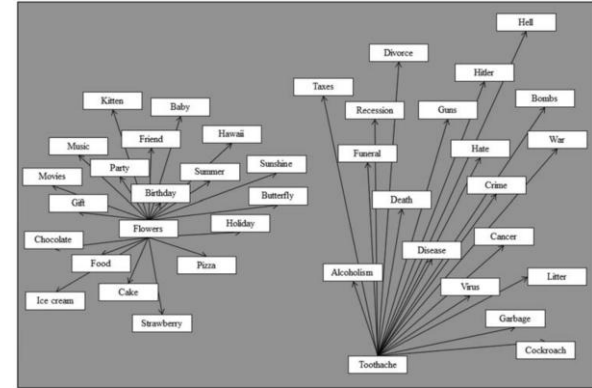
Study 5: self-selected words about
experiences rather than knowledge

A General Valence Asymmetry in Similarity: Good Is More Alike Than Bad

Alex Koch and Hans Alves
University of Cologne

Tobias Krüger
University of Heidelberg

Christian Unkelbach
University of Cologne



How General is the Similarity Asymmetry?

Facial Attractiveness

Attractive faces are more similar than unattractive faces
(Langlois & Roggman, 1990; Potter et al., 2007; Rhodes, 2006)

Personalities

Happy people are all alike (Iliev & Bennis, 2023)

Liked people are perceived to have similar personalities
(Leising et al., 2010)



Iyengar et al., 2015

How General is the Similarity Asymmetry?



Contents lists available at ScienceDirect

Journal of Experimental Social Psychology

journal homepage: www.elsevier.com/locate/jesp

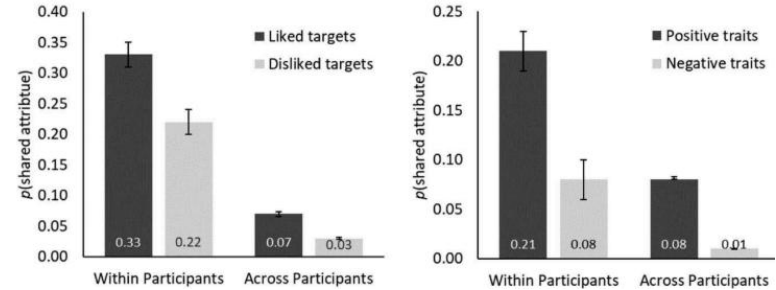
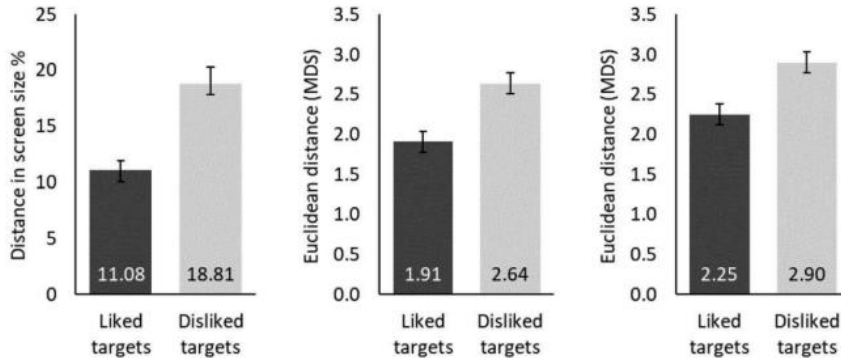
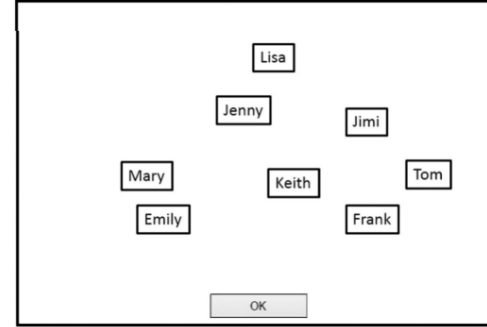


My friends are all alike – the relation between liking and perceived similarity in person perception



Hans Alves*, Alex Koch, Christian Unkelbach

University of Cologne, Germany



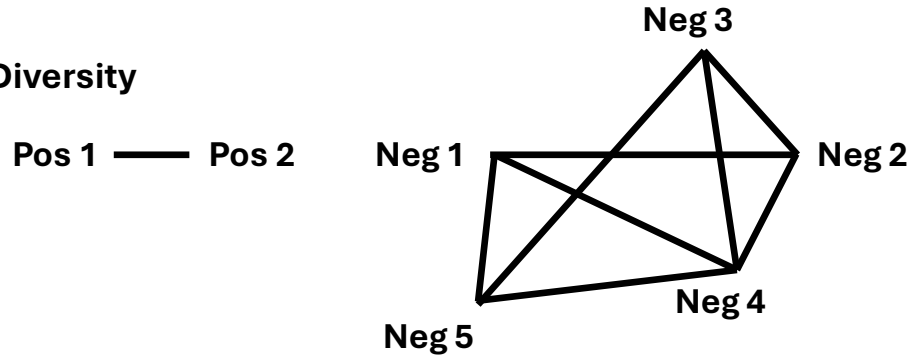
From Similarity to Diversity

Negative Information is more diverse than positive information

Similarity

Pos 1 — Pos 2 Neg 1 — Neg 2

Diversity



More negative words in vocabulary

(Leising et al., 2012; Weitzel et al., 2026)

More negative „basic“ emotions

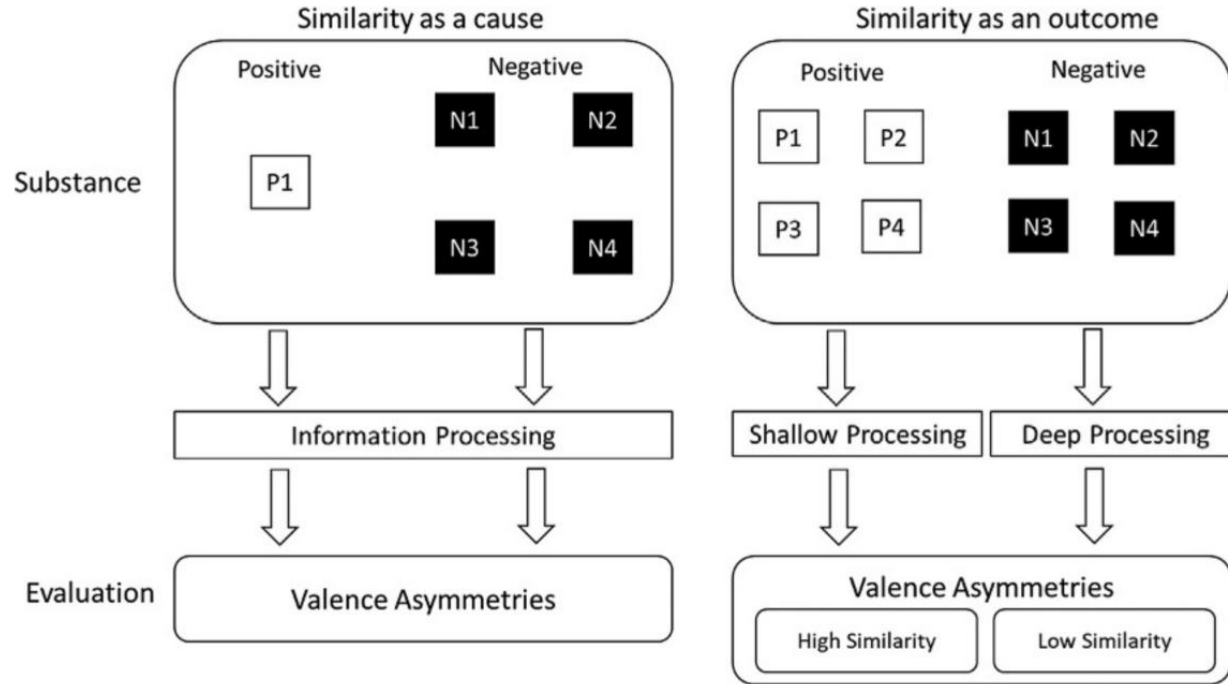
Table 2. A list of authors who enumerated basic positive and negative emotions. The columns state the named positive, negative, and ambivalent emotions and the absolute frequency of the three emotion classes within each list of "basic" emotions.

Source	Positive emotions	Number of positive emotions	Negative emotions	Number of negative emotions	Ambivalent emotions	Number of ambivalent emotions
Arnold (1960)	courage hope love	3	anger aversion dejection despair fear hate sadness	7	desire	1
Ekman, Friesen, and Ellsworth (1982)	joy	1	Anger disgust fear sadness	4	surprise	1
Gray (1982)	joy	1	rage terror anxiety	3	-	0
Izard (1971)	joy	1	anger contempt disgust distress fear guilt shame	7	interest surprise	2
James (1884)	love	1	fear grief rage anger disgust fear subjection	3	-	0
McDougall (1926)	elation tender-emotion	2	anger disgust fear	4	wonder	1
Mowrer (1960)	pleasure	1	pain	1	-	0
Ortengren and Johnson-Laird (1987)	happiness	1	anger disgust anxiety sadness	4	-	0
Panksepp (1982)	-	0	fear rage panic anger disgust fear sadness	3	expectancy	1
Plutchik (1980)	acceptance joy	2	anger disgust fear sadness	4	anticipation surprise	2
Tomkins (1984)	joy	1	anger contempt disgust distress fear shame	5	interest surprise	2
Watson (1930)	love	1	fear rage sadness	2	-	0
Weiner and Graham (1984)	happiness	1	happiness	1	-	0
Mean		1.23		3.69		0.77

Note: The complete references are in the reference list, cited after Ortengren and Turner (1990).

Explanations

Negative information's higher diversity as an „objective“ property of the external world



Explanations

Trends in Cognitive Sciences

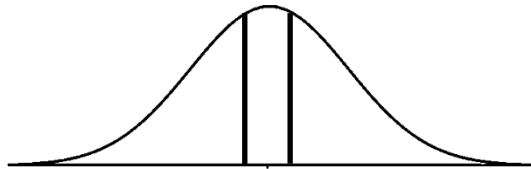
CellPress

I. The **Range Principle** (Alves et al., 2017)

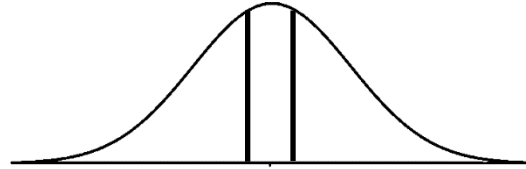
Opinion

Why Good Is More Alike Than
Bad: Processing Implications

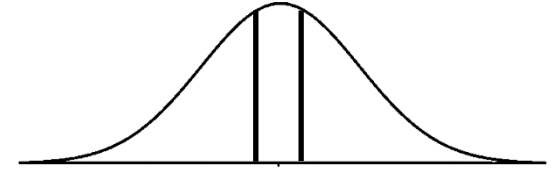
Hans Alves,^{1,2,*} Alex Koch,^{1,2,*} and Christian Unkelbach¹



Temperature



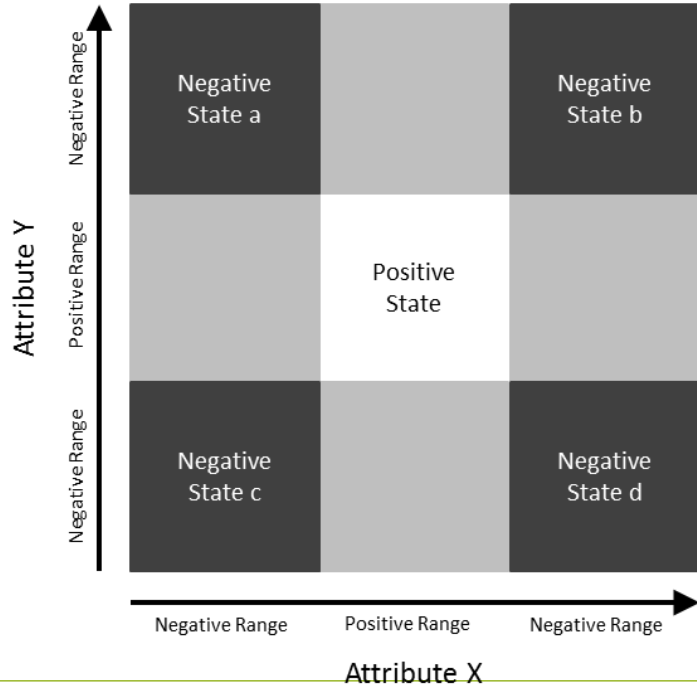
Human Height



Talking Frequency

Explanations

I. The Range Principle (Alves et al., 2017)



Object X

Positive Attributes

Negative Attributes

Trends in Cognitive Sciences

CellPress

Opinion

Why Good Is More Alike Than Bad: Processing Implications

Hans Alves,^{1,2,*} Alex Koch,^{1,2,*} and Christian Unkelbach¹

Explanations

II. The **Anna Karenina Principle** (Diamond, 1997; Moore, 2001)

Positivity requires more positive conditions than negativity requires negative conditions

“It seems that in good situations a number of requirements must hold simultaneously, while to call a situation bad even one failure suffices.” (Arnold, 1992)

Related Concept: Liebig’s Law of the Minimum:

“the worst factor determines the situation for an organism” (Gorban et al., 2010)

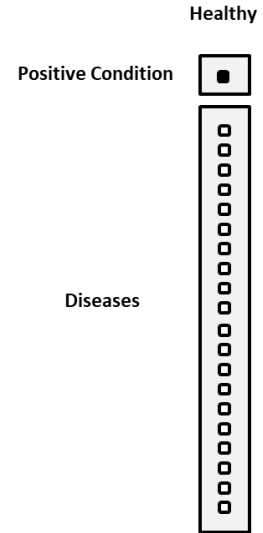
Examples:

“A deficiency in any one of a great number of factors can render a species undomesticable.” (Diamond, 1997)

Successful and unsuccessful ecological risk assessment (Moore, 2001)

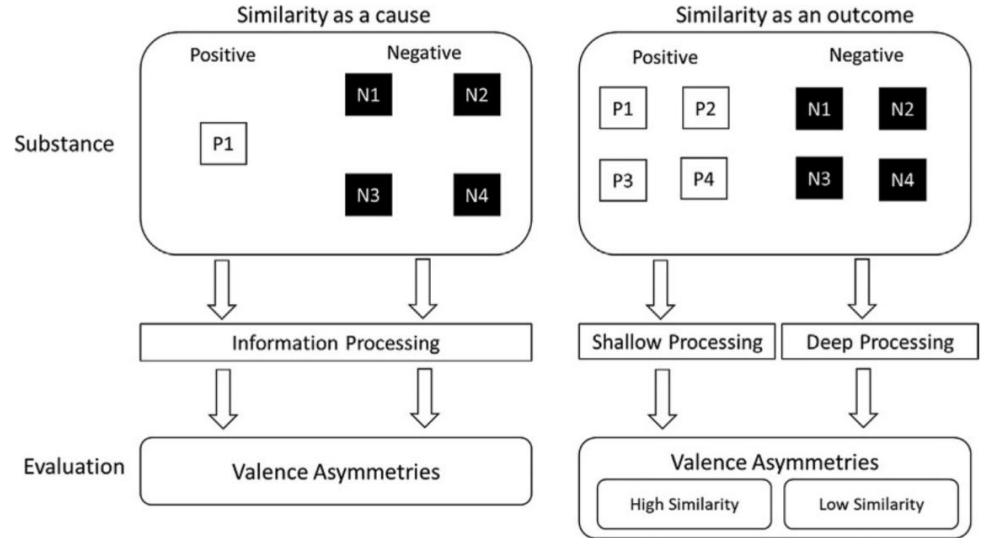
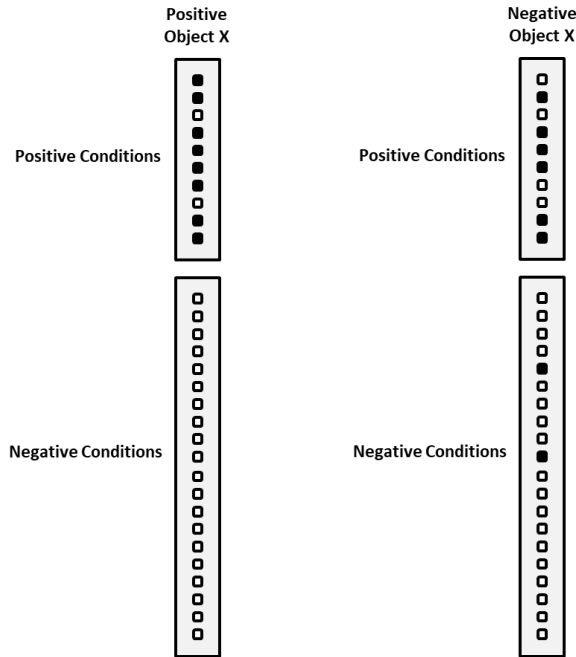
Economic systems / stock market; medicine; climate systems (Gorban et al., 2010)

Sometimes success is even defined as a the absence of negative conditions



Explanations

The **Origin** of **Valence Asymmetries**



Further Implications

Negative information is **more „interesting“** -> movies, novels, news

Negative information attracts **more attention**

Negative information can be **remembered better**

Positive information can be **processed faster**

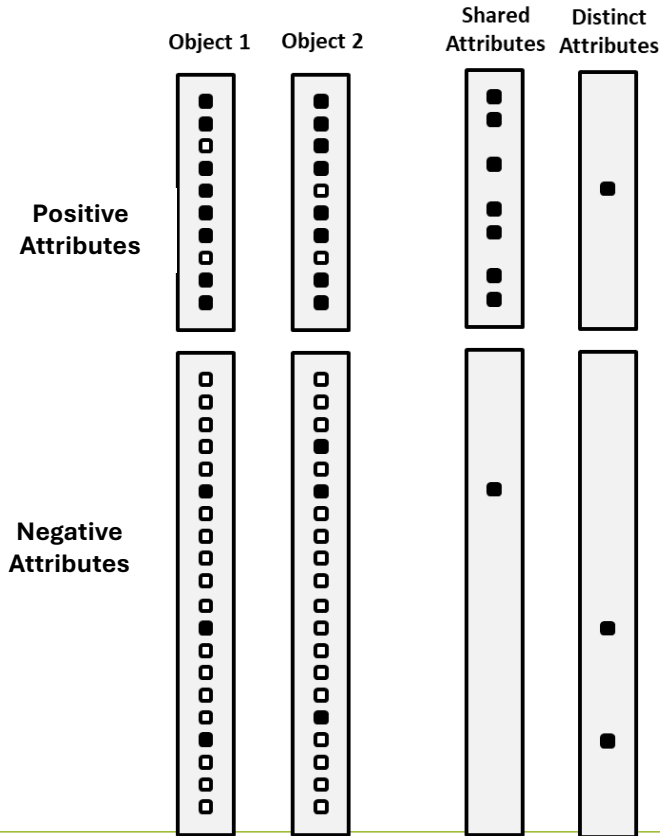
Few reasons to like, many reasons to hate

- Likes are more diagnostic than dislikes -> sharing likes attracts more (Zorn, Mata, Alves, 2022)

Negative attributes differentiate people and groups from one another while people and groups **share their positive attributes**

...

The Common Good Phenomenon



Journal of Experimental Psychology: General
2017, Vol. 146, No. 4, 512–528

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0096-3445/17/\$12.00 http://dx.doi.org/10.1037/xge0000276

The “Common Good” Phenomenon: Why Similarities Are Positive and Differences Are Negative

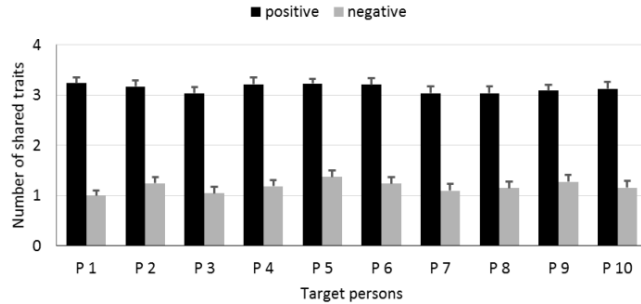
Hans Alves, Alex Koch, and Christian Unkelbach
University of Cologne

1. Negative attributes are more **diverse** than positive attributes
e.g. faces, traits, individuals, words, emotions, etc.
(e.g. Alves et al., 2015, M&C; Alves et al., 2016, JESP; Alves et al., 2017, TICS; Alves et al., 2018, C&E)
2. Positive attributes are more **frequent** than negative attributes
(e.g. Alves et al., 2017, Jep:G; Rothbart & Park, 1986; Matlin & Stang, 1978; Wood & Furr, 2016)

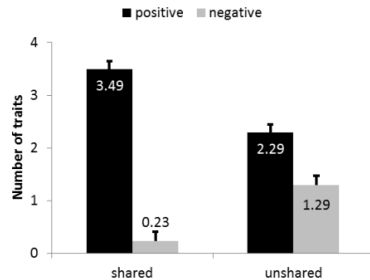
Negative attributes = distinct
Distinct attributes = negative

The Common Good Phenomenon

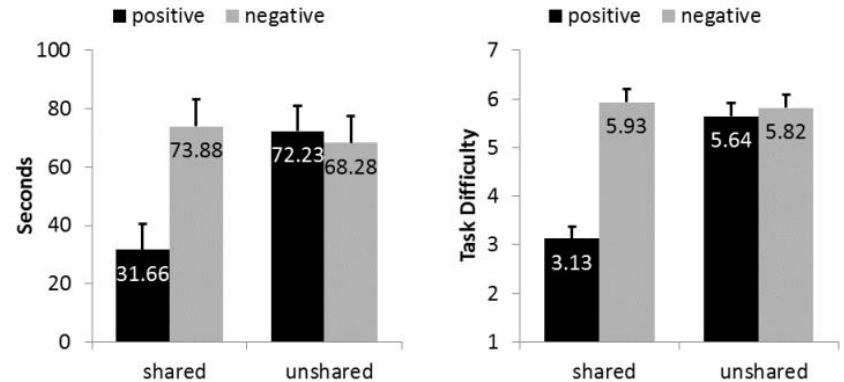
The probability of positive and negative traits being shared by other people.



The probability of shared and distinct traits being positive/negative



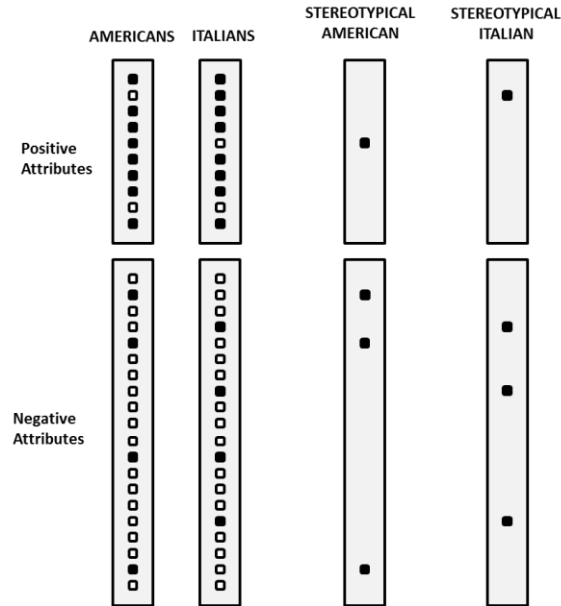
Accessibility of shared/distinct positive/negative attributes



Implications for Perception of Groups

Most stereotypes are negative

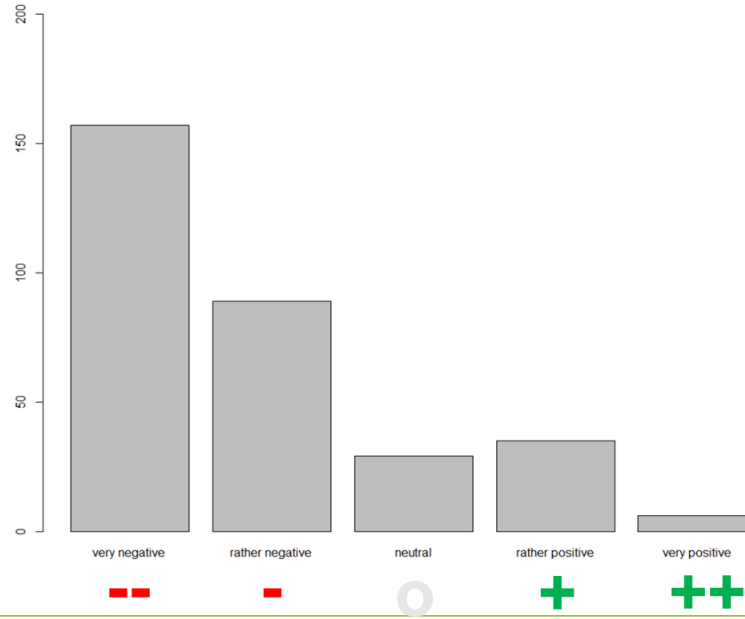
Stereotypes = Differentiating Attributes = Negative Attributes



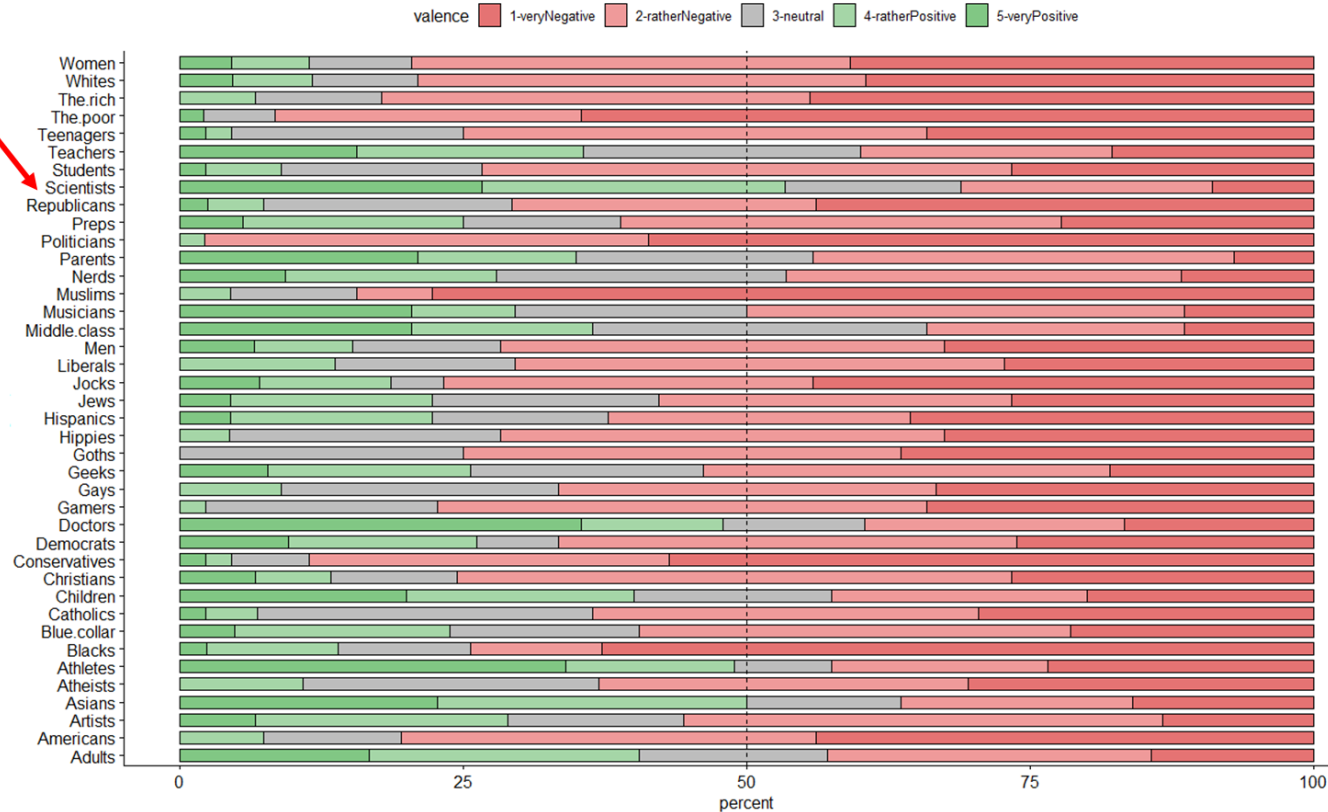
Negative Stereotypes

Why most stereotypes are negative (Unkelbach, Weitzel, & Alves, under review)

“Please provide two attributes used in society and the media to describe social groups”.

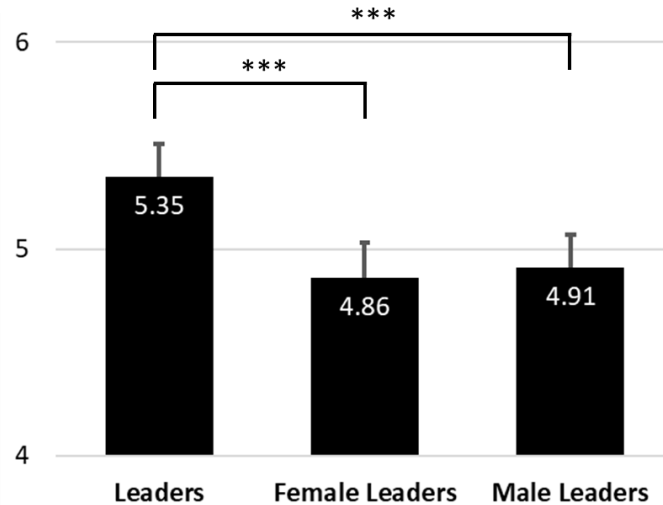


Negative Stereotypes



Negative Stereotypes

Paradoxical Consequences: Exhaustive sub-groups are perceived more negatively than their superordinate groups (Schulte, Weitzel, Alves, & Lammers, 2026; SPPS)

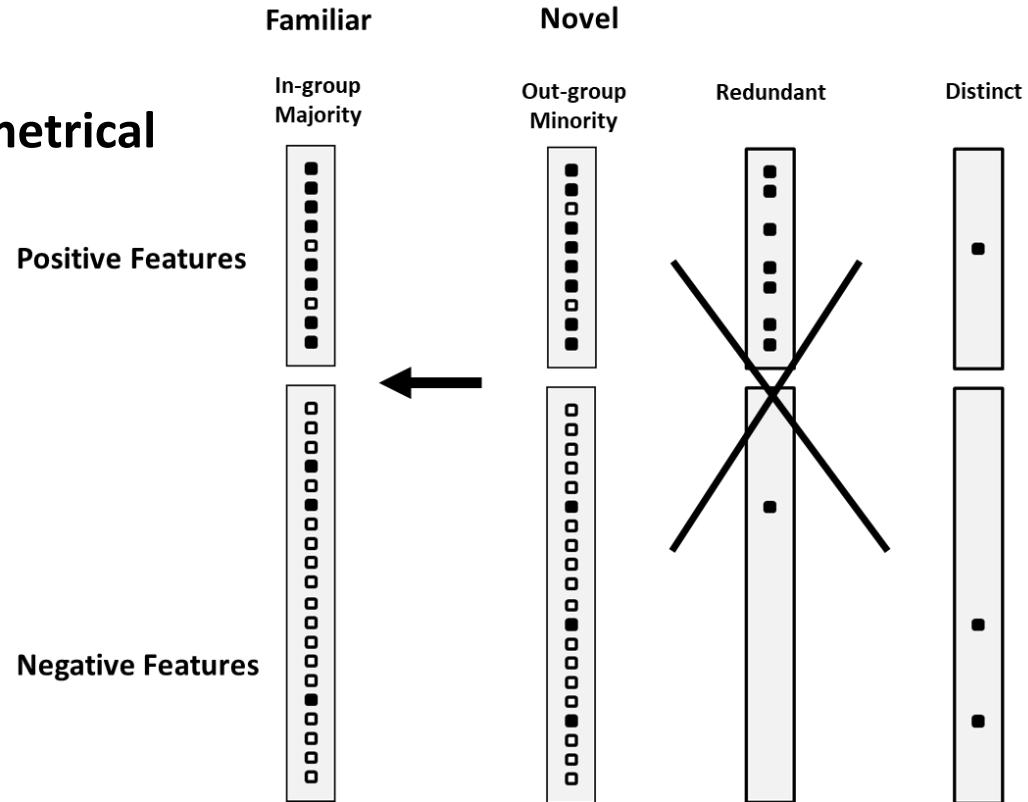


$t(279) = 3.31, p < .001, d = .23$

$t(279) = 3.58, p < .001, d = .25$

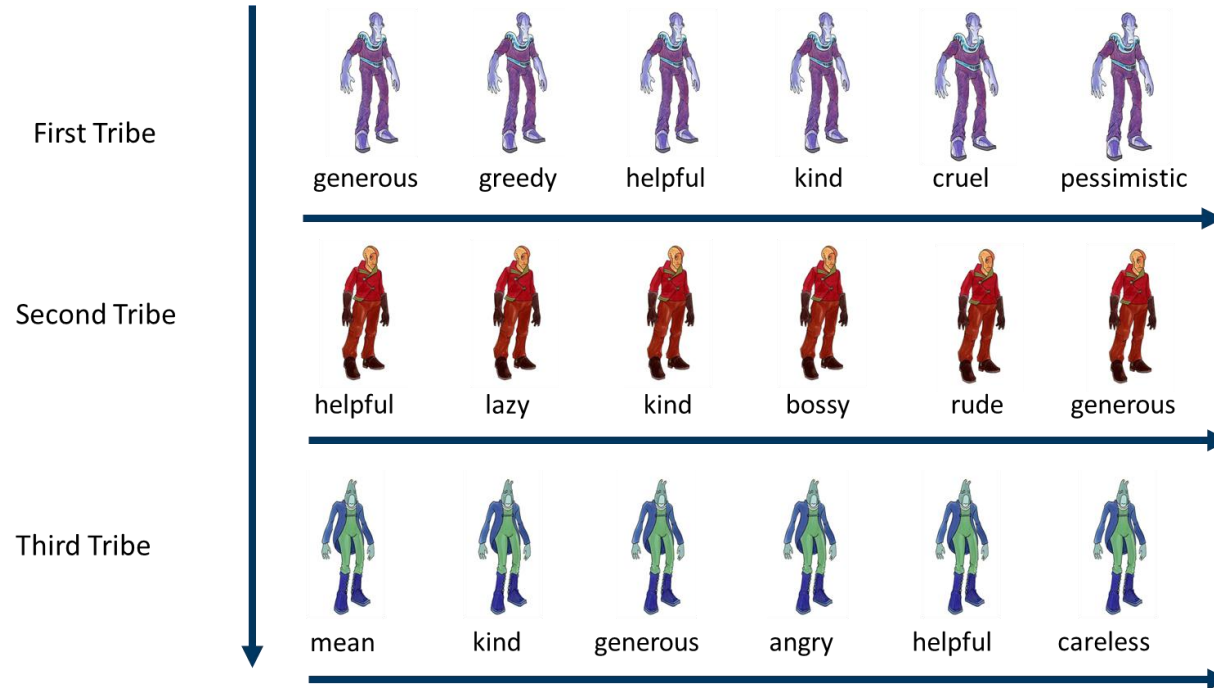
Differentiation and Intergroup Biases

Differentiation is **asymmetrical**



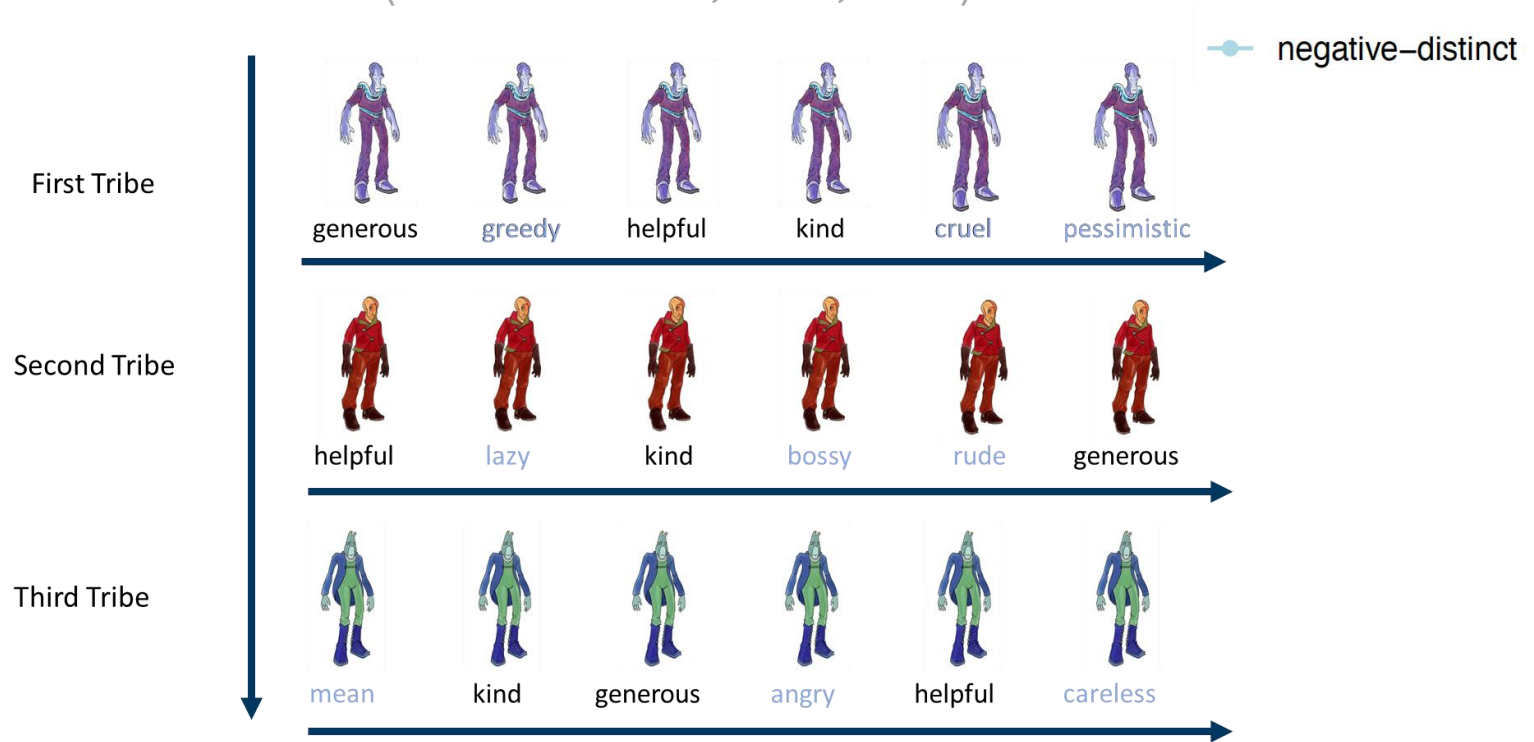
Differentiation and Intergroup Biases

(Woitzel & Alves, 2024, PSCI)



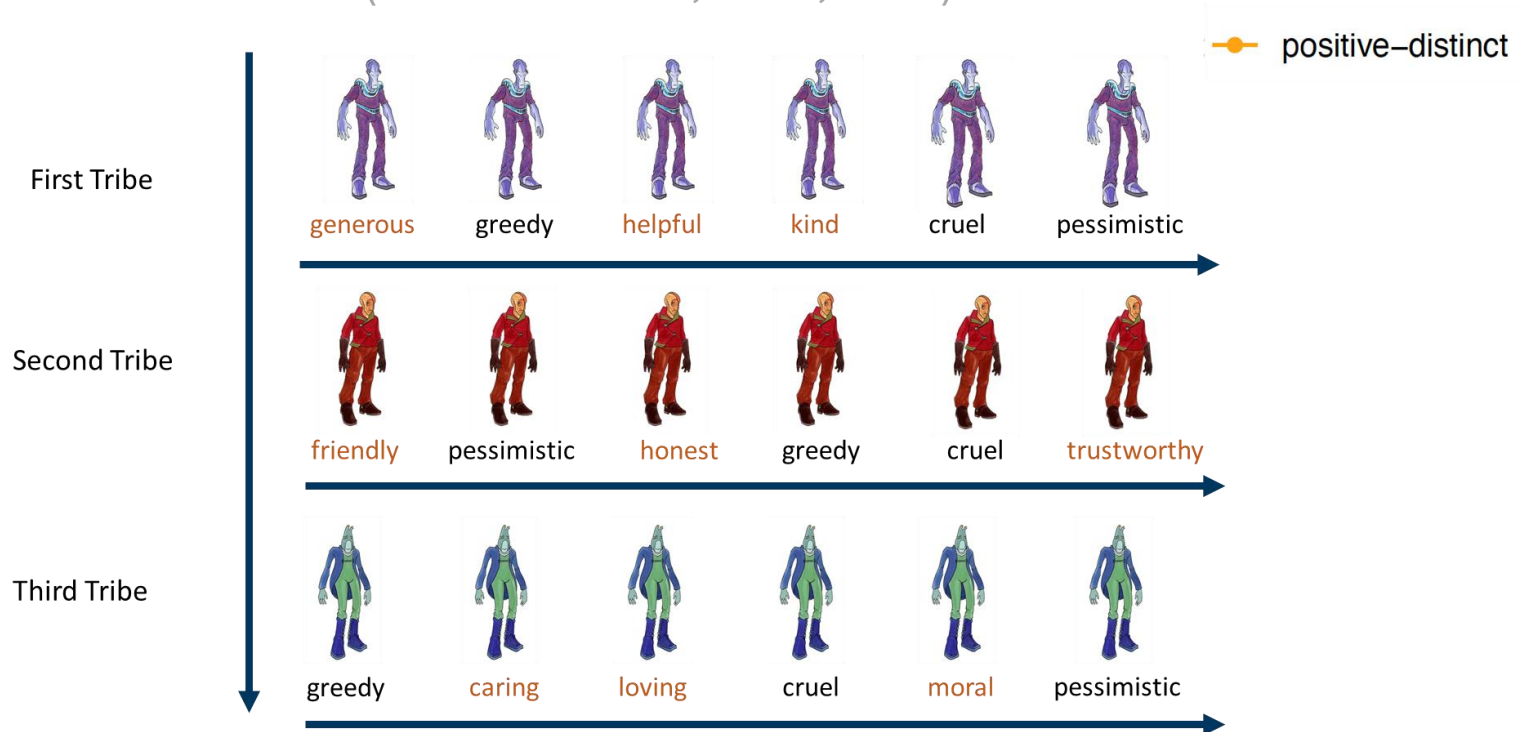
Differentiation and Intergroup Biases

(Woitzel & Alves, 2024, PSCI)



Differentiation and Intergroup Biases

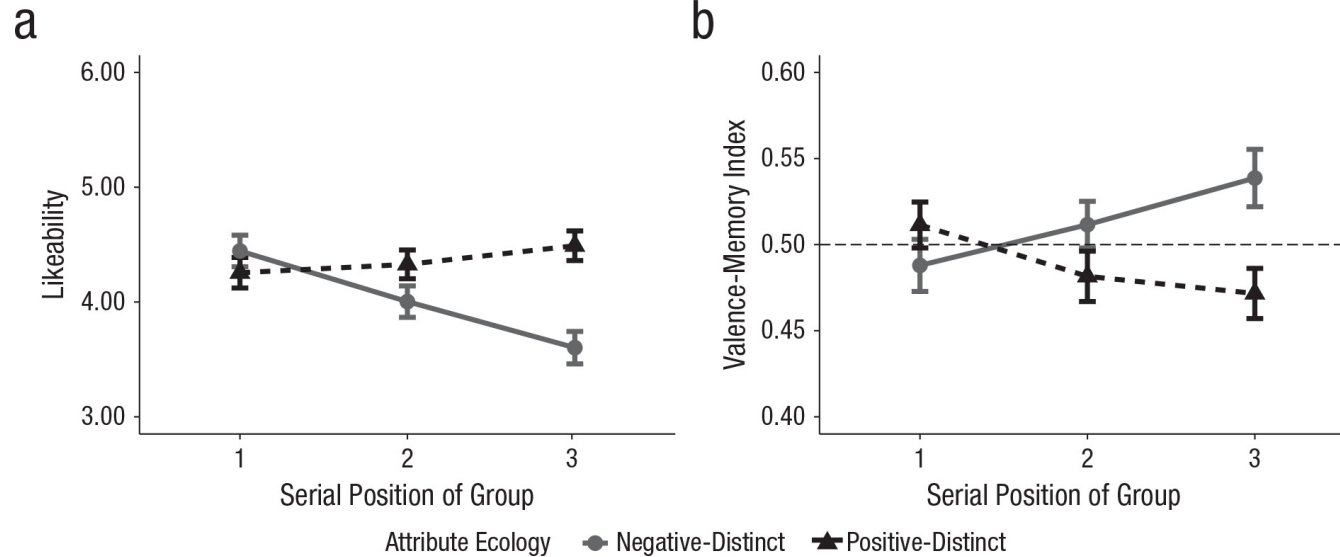
(Woitzel & Alves, 2024, PSCI)



17

Differentiation and Intergroup Biases

(Woitzel & Alves, 2024, PSCI)



Conclusions

- Valence asymmetries are exciting because they are related to everything that humans care about
- There are many different valence asymmetries with different causes
- Some of these causes may be objective properties of the external world
- Some may be inside our „heads“

Thank You!



Johanna Woitzel



Anna Schulte



Moritz Ingendahl



Tabea Zorn



Christian Unkelbach



Anne Weitzel



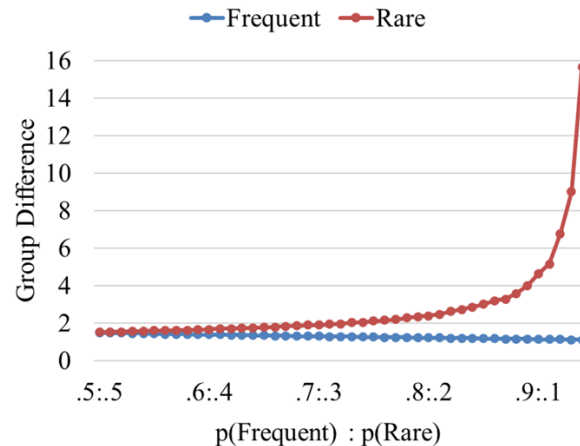
Alex Koch

Differentiation and Negative Stereotypes

Explanation:

- Stereotypes = Attributes that differentiate groups; diagnostic for group membership
➔ maximize $p(\text{group} | \text{attribute})$ and not $p(\text{attribute} | \text{group})$!
- Rare attitudes are more diagnostic (for group membership)
➔ Negative attributes are more diagnostic

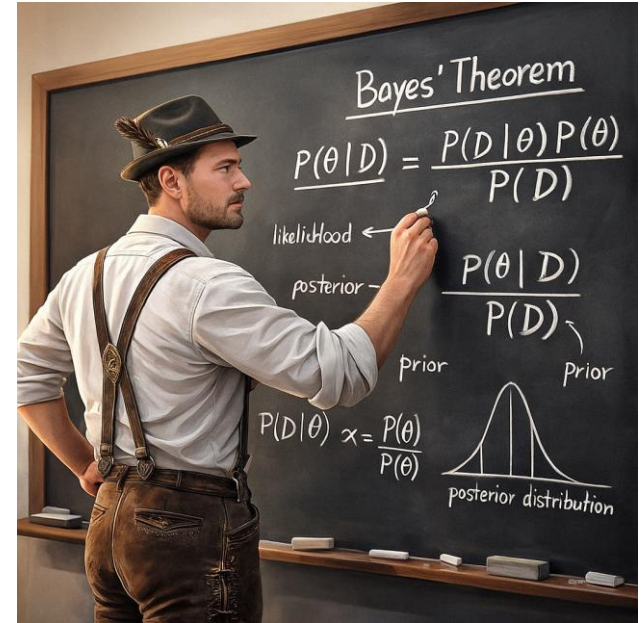
	Female	Male	Relative Likelihood	
Not Harassed	4800	4950	1.03	9750
Harassed				250
	5000	5000		10000



Differentiation and Negative Stereotypes

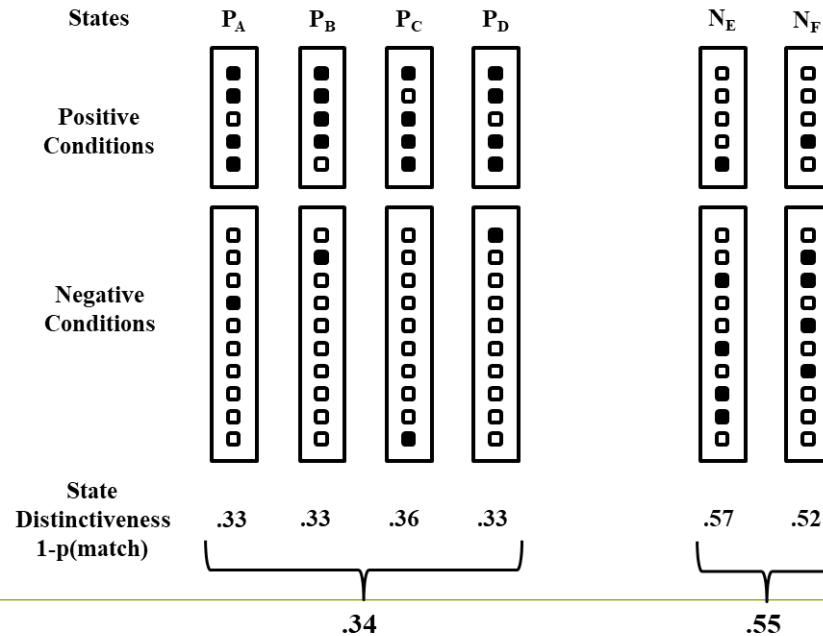
Example Lederhosen:

- $p(\text{German}) = 0.05$
- $p(\text{Lederhosen} | \text{German}) = .10$
- $P(\text{Lederhosen} | \text{non-German}) = .002$
- ➔ $P(\text{German} | \text{Lederhosen}) = .72$; 2.6 times more likely German than non-German



Good / Bad – Success / Failure

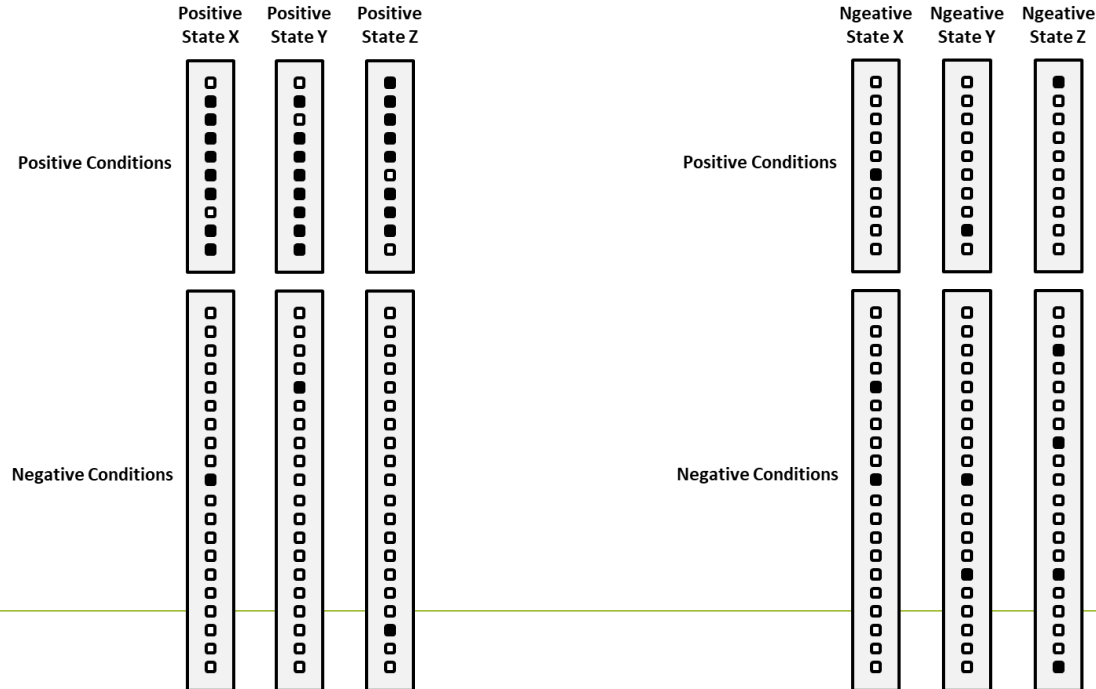
Implication: States of failure are more diverse than states of success



Good / Bad – Success / Failure

Implication for interest / curiosity: Priority of failure, misery, bad characters

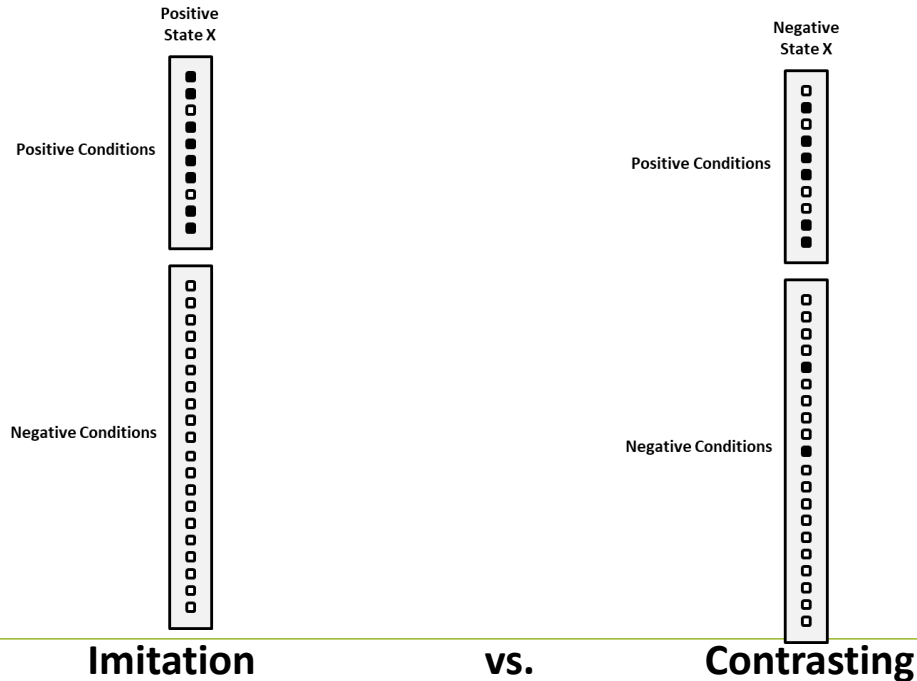
Alves et al., 2023; Wylie & Gantman, in press. „People are curious about immorality and agents who differ from the norm”



Good / Bad – Success / Failure

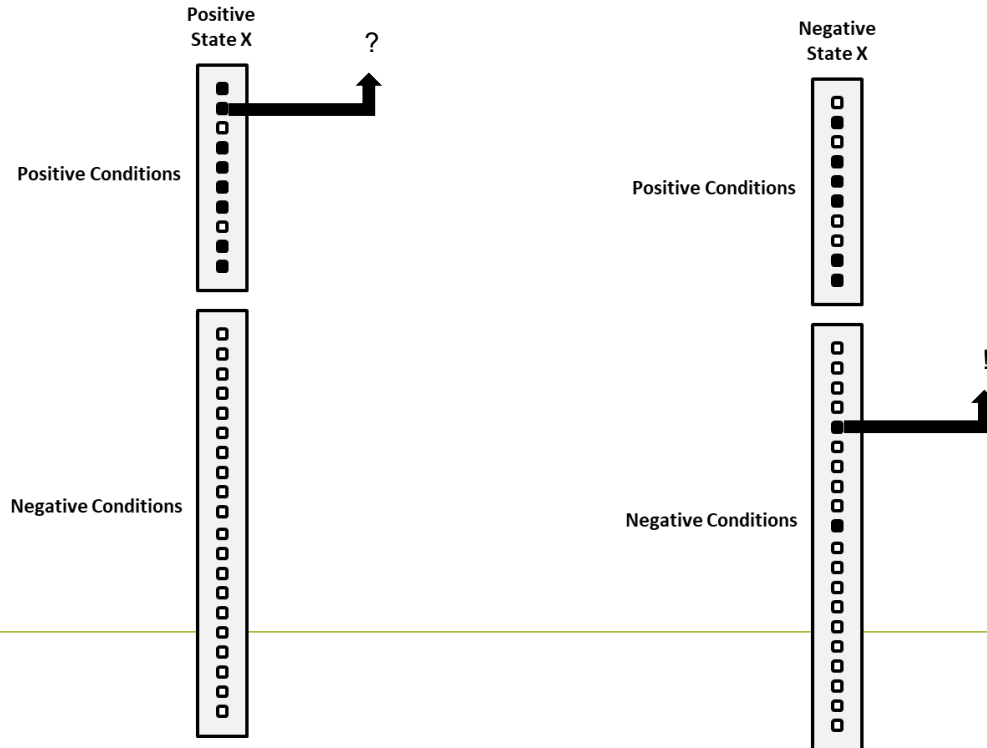
Implication for learning: More effective to learn from positive than negative exemplars

Counter to Eskreis-Winkler & Fishbach (2020)



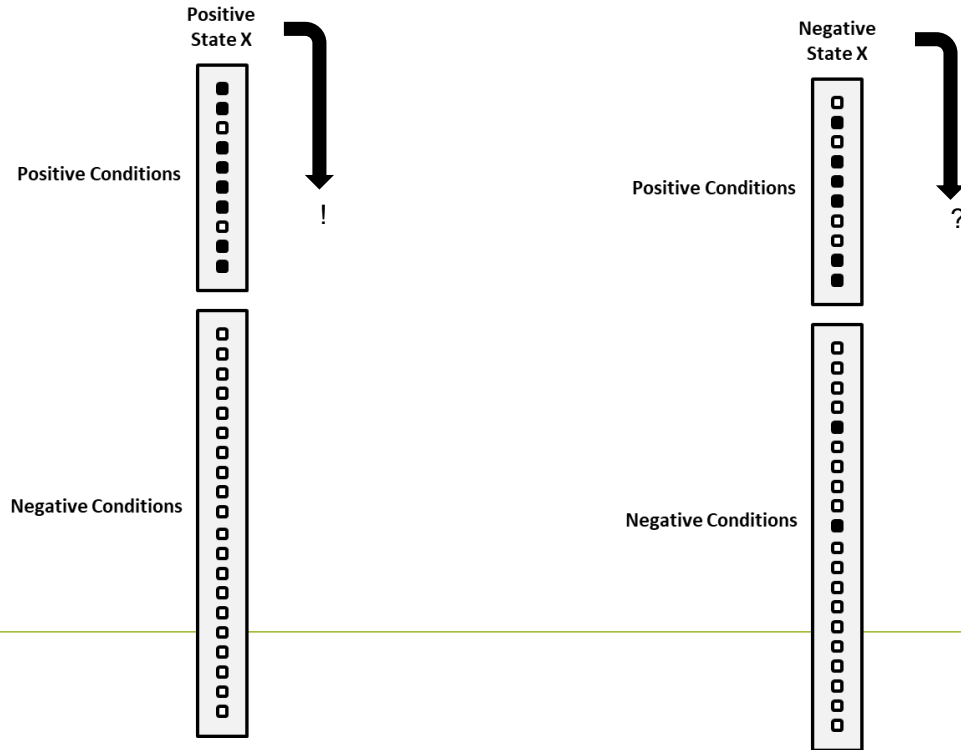
Good / Bad – Success / Failure

Implication for inferences: Negative conditions are more diagnostic for outcome valence than positive conditions (Skowronski & Carlston, 1987)



Good / Bad – Success / Failure

Implication for inferences: Positive outcomes are more diagnostic for its underlying conditions than negative outcomes



Good / Bad – Success / Failure

Implication for inferences: Positive outcomes are more diagnostic for its underlying conditions than negative outcomes

Example from communication:

- How are you? How was it? How did you like it?
 - Good
- How are you? How was it? How did you like it?
 - Not great
- Why?

Good / Bad – Success / Failure

Implication for inferences: Positive outcomes are more diagnostic for its underlying conditions than negative outcomes

Example from Impression Formation (Zorn, Mata, Unkelbach, & Alves, in prep.)

1.

„Please imagine you meet a person and you learn that this person **likes** the movie *Avatar*.“

attitude domain

example

IV: valence

like vs. dislike
(within-
participants)



„Please imagine you meet a person and you learn that this person **dislikes** the movie *Inception*.“

attitude domain

example

8x

2.

DV: anticipated knowledge about another person

How well do you think you could anticipate this person's attitudes?

not at all



very well



How well do you think you could anticipate this person's personality?

not at all



very well



How well do you think you could anticipate who this person is?

not at all



very well

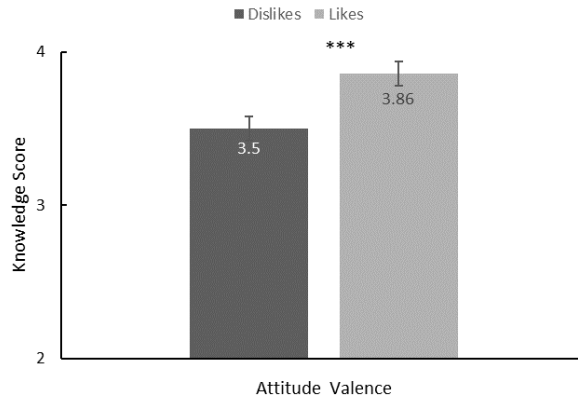


mean = composite
knowledge score (**DV**)

Good / Bad – Success / Failure

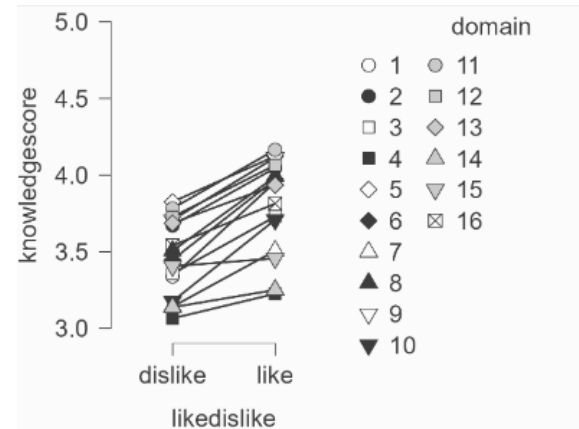
Implication for inferences: Positive outcomes are more diagnostic for its underlying conditions than negative outcomes

Example from Impression Formation (Zorn, Mata, Unkelbach, & Alves, in prep.)



.. more knowledge based on likes
(replicated basic effect)

$t(198) = 7.54, p < .001, d_z = 0.53$



$b = 0.18, SE = 0.02, t(18.15) = 7.49, p < .001$